

Source-First Growing-Shell Stability of the Native Weighted/Positive Budget Gap

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Abstract

The preceding finite native-profile audit found a robust budget separation between a signed weighted target and an all-positive control, but its target labels were available on only 18 inherited rows. We extend the hostile test to the 34-row growing/control grid of the prime-shell dynamical model and recompute the weighted target from the physical output Gram matrix on every row. The finite Gram and sign-enumeration identities are exact. At relative RMS tolerances $1/4$, $1/2$, and $3/4$, the common-prefix weighted/positive budget gap exceeds 10 on all 34 rows; its minima are 85.3204, 38.2187, and 39.2637. The common weighted budget exceeds 10^{-5} in all 102 row-tolerance cases under each of three source normalizations. This is a source-first finite growing-grid certificate. It does not prove uniform profile-budget growth, arithmetic L^2 , or the twin-prime conjecture.

1 Question and finite model

TPC-301 showed that a finite native obstruction survives a tolerance ladder, a common-prefix comparison, and three source normalizations [1]. The unresolved question was whether that obstruction depended on the inherited 18-row target atlas. Here the target itself is regenerated on the complete TPC-288 growth/control grid.

Let S be a finite prime shell and let g_q be the exact literal physical output vector attached to $q \in S$. Put

$$G_{q,r} = \langle g_q, g_r \rangle, \quad R(a) = \frac{a^T G a}{\text{tr } G}, \quad a_q \in \{\pm 1\}.$$

The weighted target a_w is the exact minimizer of R after fixing one global sign; the positive control is $a_+ = (1, \dots, 1)$. The source profiles are the first 17 literal cutoff profiles

$$u_z(t) = \lambda(t) - \sum_{d \leq z, d|t} \mu(d), \quad z = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61.$$

For the first k profiles, write $V_k = A^T U_k$ and $M_k = U_k^T U_k$. We use the relative target budget

$$B_{k,\tau}(b) = \min\{c^T M_k c : \|V_k c - b\|_2 \leq \tau \|b\|_2\}. \quad (1)$$

The grid has 16 growth-path rows and 18 source-control rows. Their shell lists contain 430 explicit targets. The parent metadata also contains a 1,380-edge count for a different inherited census; we keep the two numbers separate.

Table 1: Minimum weighted/positive budget gap over the 34 rows.

relative RMS τ	common prefix	full prefix	common > 10
1/4	85.3203517096	85.3203517096	34/34
1/2	38.2186652435	38.2186652435	34/34
3/4	39.2637006403	26.0731501545	34/34

2 Exact finite identities

Proposition 1 (Gram identity and global signs). *For every finite shell, G is positive semidefinite and $R(a) = R(-a)$. Fixing the first sign to $+1$ therefore represents every global-sign class.*

Proof. For any coefficient vector c ,

$$c^T G c = \sum_{q,r} c_q c_r \langle g_q, g_r \rangle = \left\| \sum_q c_q g_q \right\|_2^2 \geq 0.$$

The quadratic form contains each sign twice, so replacing every sign by its negative leaves it unchanged. $\square$

Proposition 2 (Finite enumeration). *After clearing a common positive denominator, a reflected Gray traversal of the $2^{|S|-1}$ tail bit strings gives the exact minimum of R on the equal-sign domain.*

Proof. The traversal flips one tail coordinate at a time and visits every tail bit string once. The incremental quadratic update is an algebraic rearrangement of the direct integer quadratic form, so no floating-point comparison is used in selecting the minimum. $\square$

Proposition 3 (Budget monotonicity and normalization). *Relaxing τ or enlarging a profile prefix cannot increase (1). At a common prefix, a positive target-independent source normalizer cancels from the weighted/positive budget ratio.*

Proof. Both statements follow by inclusion of feasible sets. For normalization, divide both numerator and denominator by the same positive scalar. $\square$

3 Source-first audit

For every row we rebuild the source weights and all physical output vectors with rational arithmetic, form G , and enumerate $2^{|S|-1}$ sign classes. The resulting a_w is then inserted into (1), alongside a_+ . At each $\tau \in \{1/4, 1/2, 3/4\}$ we retain three contexts: the smallest common feasible prefix, target-specific first feasible prefixes, and the full prefix. The primary comparison is the common-prefix ratio. The source normalizers are $\|\beta\|_2^2$, $\text{tr}(M_k)/k$, and $M_k[1, 1]$.

The frontier is solved by 60-digit ridge-parameter bisection. Decimal intervals are written outward; exact rational labels and ratios are retained in the canonical certificate. An independent checker rebuilds all 34 Gram matrices without importing this producer.

4 Results

All 34 source-first weighted ratios are below one and all 34 positive ratios are above one. The common weighted budget is above 10^{-5} in 102/102 cases for each source normalizer. The common-prefix normalization identity is replayed in 102 cases, and full-prefix tolerance monotonicity in

68 target-row checks. These counts include the declared source controls and do not mean that all admissible profiles or shells have been tested.

The finite result is stronger than a mere reuse of the old target atlas: the 17-prime endpoint row, for example, is assigned its label by enumerating all 2^{16} global-sign classes on its own physical Gram matrix. The result is still only a stability map. In particular, the displayed minima do not constitute a lower bound uniform in shell size, source scale, or profile cutoff.

5 Claim firewall and next question

The exact claims are finite Gram positivity, global-sign reduction, exhaustive enumeration, budget monotonicity, and normalization cancellation. The atlas claims only the 34 declared rows, 430 explicit shell targets, 17 profiles, three tolerances, and two target classes. A uniform native profile-budget theorem and literal arithmetic L^2 estimate remain open; the fixed-power credit is zero and no full Gate-B or twin-prime result is claimed. The natural next project is to test whether the native budget has a uniform growth law, or to construct the first growing-shell counterexample.

Reproducibility

The repository contains the canonical certificate, an independent source-first replay, exact theorem stress fixtures, and a fail-closed local Bridge-B checker. The Session-named Route-A/Route-B evaluator files are not present in this checkout, so no official evaluator pass is asserted.

References

- [1] Liang Wang. Tolerance and source-normalization robustness of the native budget gap, 2026. TPC-301 project release in the accompanying repository.